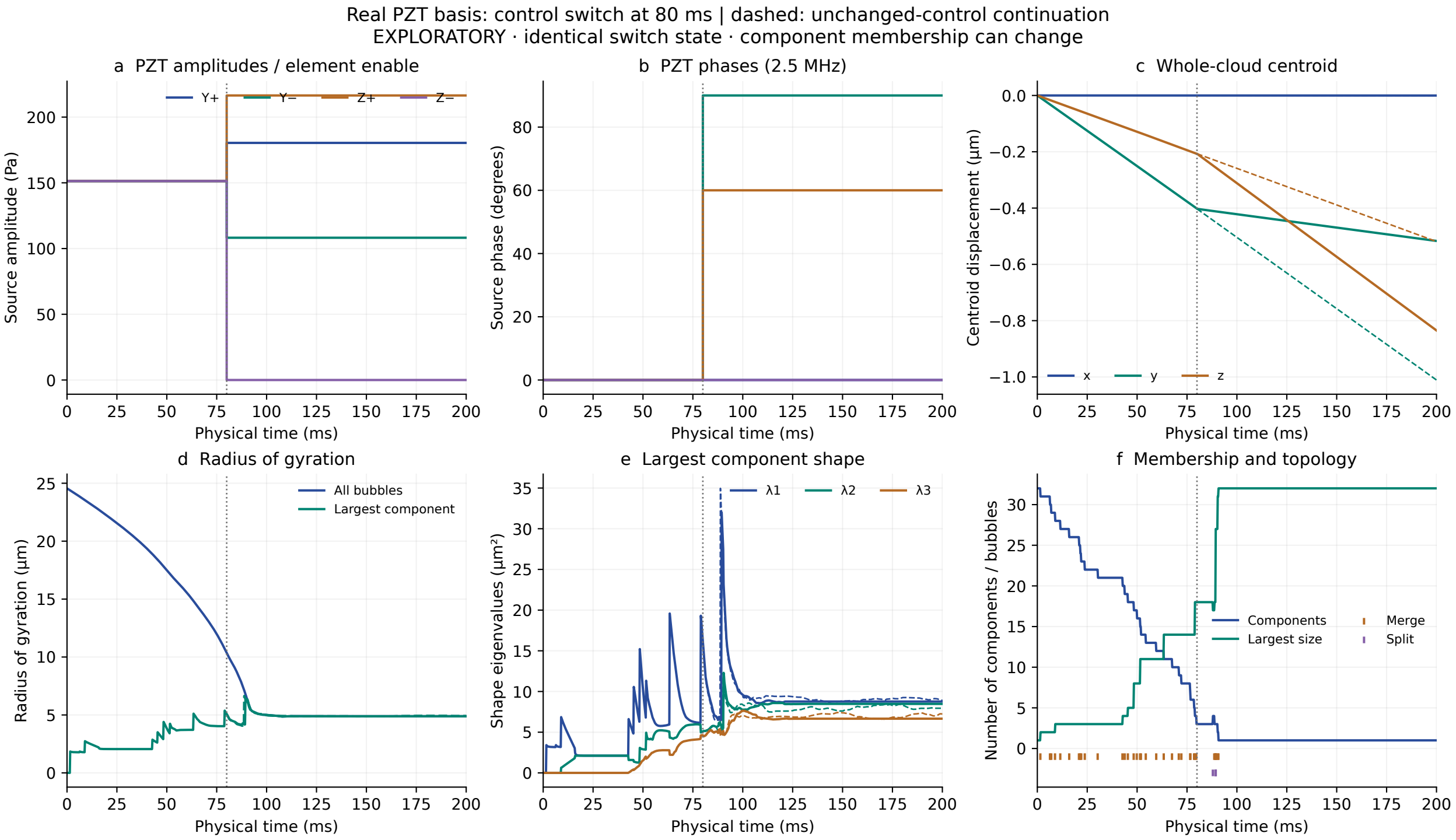
Real PZT basis: control switch at 80 ms | dashed: unchanged-control continuation
EXPLORATORY · identical switch state · component membership can change
a PZT amplitudes / element enable
Y+
Y−
Z+
Z−
Source amplitude (Pa)
Physical time (ms)
b PZT phases (2.5 MHz)
Source phase (degrees)
Physical time (ms)
c Whole-cloud centroid
Centroid displacement (µm)
x
y
z
Physical time (ms)
d Radius of gyration
All bubbles
Largest component
Radius of gyration (µm)
Physical time (ms)
e Largest component shape
λ1
λ2
λ3
Shape eigenvalues (µm²)
Physical time (ms)
f Membership and topology
Components
Largest size
Merge
Split
Number of components / bubbles
Physical time (ms)